

MY APPLICATION BOT

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THIS IS ABOUT THE APPLICATION STATERGIES DATA

Based on everything you've shown, **22A and 22B are not fully complete from a profitability perspective**, but they are complete from a **code implementation perspective**.

The biggest finding is:

Current State

- ✓ Signal bottlenecks mostly removed (22A)
- ✓ Real metrics flow into Risk V2 (22B)
- ✓ Dynamic sizing now works
- ✓ Drawdown scaling works
- ✓ Winning strategies receive more capital
- ✓ Losing strategies get penalized
- ✗ Kelly sizing is still practically ineffective
- ✗ No statistically validated profitable strategy exists yet
- ✗ No 500–1000 trade evidence exists
- ✗ Institutional alpha engines are still the largest unrealized edge

What Should Be Done Next

Priority 1 — Phase 22C

This is now the highest ROI task.

Your own audit found:

- Funding Mean Reversion
- Liquidation Cascade
- Liquidity Sweep
- FVG
- Order Block
- MSS
- Session Expansion

are the strongest alpha sources in the entire codebase.

But most are:

- not dispatched
- not fed data
- not generating signals

Current state:

Category	Status
Indicator strategies	Running
EMA variants	Running
RSI variants	Running
Alpha engines	Mostly dormant

Meaning:

The application is currently trading the weakest 80% of its strategy inventory.

Priority 2 — Phase 22D

After alpha is active:

Measure:

- Fill quality
- Missed entries
- Latency
- TP overrides
- Bridge delays

- Slippage

Without this:

Good signals can still become bad trades.

Priority 3 — Massive Validation Campaign

This is where profitability is actually proven.

Run:

Validation 1

Top 20 strategies

Target:

- 500 trades each
-

Validation 2

Top 10 strategies

Target:

- 1000 trades each
-

Collect:

- Win Rate
 - Profit Factor
 - Sharpe
 - Sortino
 - Expectancy
 - Max DD
 - Regime performance
-

What You Need To Know

Right now:

You have evidence that:

- the architecture works
- the risk engine works
- sizing works
- allocation works

You do NOT yet have evidence that:

- strategy edge exists

Those are completely different things.

A perfect risk engine on a bad strategy still loses money.

A mediocre risk engine on a good strategy can make money.

My Assessment

If your audits are accurate:

Current Profitability Potential

Before 22A:

~2/10

After 22A + 22B:

~5/10

After 22C:

~8/10

After 22D:

~8.5/10

After 1000-trade validation:

Unknown — this is where the truth appears.

The Most Important Next Question For Claude

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Perform a Phase 22C Institutional Alpha Activation Audit.

Do not modify code.

Analyze the entire codebase and produce a forensic report covering:

1. Every alpha engine currently implemented.
2. Which alpha engines generate signals today.
3. Which alpha engines are dormant.
4. Exact reasons dormant modules are not producing signals.
5. Missing OnCandle dispatch paths.
6. Missing funding feeds.
7. Missing liquidation feeds.
8. Missing order book feeds.
9. Missing market structure feeds (FVG, MSS, CHOCH, BOS, Order Blocks).
10. Exact files and line numbers blocking activation.
11. Estimated signal frequency per alpha module after activation.
12. Expected contribution to trade count.
13. Expected contribution to edge quality.
14. Rank all alpha modules by expected profitability potential after activation.
15. Produce a detailed implementation roadmap to activate all institutional alpha modules with minimal risk and no architectural regressions.

Use only code evidence. Do not estimate profitability without measurable proof. Distinguish clearly between implemented, partially implemented, and fully operational alpha systems.

That report will tell you whether the application already contains enough profitable alpha or whether entirely new strategies need to be built.

What Should Be Done Next

Priority 1 — Phase 22C

This is now the highest ROI task.

Your own audit found:

Funding Mean Reversion

Liquidation Cascade

Liquidity Sweep

Copy and paste this directly into Claude Agent:



PHASE 22C + 22D + 22E MASTER PROFITABILITY AUDIT & IMPLEMENTATION PROGRAM

Act as a Chief Quantitative Research Officer, Hedge Fund Portfolio Manager, Institutional Market Microstructure Specialist, Senior Trading Systems Architect, Risk Director, Execution Engineer, and Lead Software Auditor.

Your mission is to determine whether this application contains genuine profitable trading edge and, if not, transform it into a statistically validated, production-grade profitable trading system.

You must perform a complete end-to-end forensic analysis of the entire codebase.

Do NOT provide assumptions.

Do NOT provide theoretical answers.

Use only code evidence, actual implementation evidence, execution paths, data flows, signal flows, strategy logic, and statistical evidence.

Audit every alpha engine in the system.

Specifically inspect:

- Funding Mean Reversion
- Liquidation Cascade
- Liquidity Sweep
- Fair Value Gap (FVG)
- Order Block
- Market Structure Shift (MSS)
- CHOCH
- BOS
- Session Expansion
- Delta Absorption
- CVD Divergence
- POC Bounce
- Order Flow Pressure

- All Phase 11 Microstructure Modules
- Any Smart Money Concepts implementation

For every alpha module provide:

1. Current status
 - Fully operational
 - Partially operational
 - Dormant
 - Dead code
2. Signal generation path
3. Data dependencies
4. Execution path
5. Missing wiring
6. Missing dispatch logic
7. Missing feeds
8. Missing collectors
9. Missing storage
10. Missing event propagation
11. Missing market structure generation
12. Exact files
13. Exact functions
14. Exact line numbers
15. Exact code fixes required

Produce a table:

Module | Status | Signal Path | Missing Components | Effort | Expected Impact

Trace all live market data.

Verify:

Price

Volume

Funding

Liquidations

Order Book

Trades

Delta

CVD

Market Profile

Volume Profile

Session Data

For each feed determine:

- Source
- Collector
- Cache
- Processing pipeline
- Consumers
- Signal usage

Identify every feed that is:

- Missing
- Stubbed
- Returning zero values
- Returning placeholder values
- Never consumed

Provide exact implementation roadmap.

Audit every strategy.

Current registry approximately 600+ strategies.

Classify into:

A. Institutional Alpha

B. Smart Money Concepts

C. Market Microstructure

D. Statistical Arbitrage

E. Mean Reversion

F. Trend Following

G. Momentum

H. Indicator-Based

I. Expansion Pack Variants

J. Duplicate Parameter Variants

Determine:

- Unique strategy concepts
- Duplicates
- Near duplicates
- Parameter clones
- Overfit candidates

Produce:

Top 50 strategies ranked by actual code quality.

For each provide:

- Alpha source
- Complexity score
- Edge potential
- Data requirements
- Robustness score
- Overfit risk
- Production readiness

Rank all strategy families by expected profitability potential.

Use:

Signal quality

Market structure awareness

Order flow awareness

Microstructure awareness

Adaptive logic

Risk integration

Produce:

Rank | Family | Edge Potential | Confidence

Determine:

Which strategies deserve capital.

Which should be retired.

Which should be merged.

Which should be rewritten.

Audit entire execution pipeline.

Trace:

Signal

- Aggregator
- Risk Engine
- Position Engine
- OMS
- Executor
- Fill

Measure:

1. Latency sources
2. Signal delays
3. Bridge parking
4. Approval delays
5. Expired signals
6. Missed entries
7. Slippage
8. TP overrides
9. Stop overrides
10. Position sizing changes
11. Execution quality degradation

For every bottleneck provide:

- Exact code location
- Root cause
- Severity
- Expected profitability impact
- Fix recommendation

Determine whether the system is capable of proving profitability.

Verify:

Trade persistence
Paper trade persistence
Backtest persistence
Walk-forward persistence
Event sourcing
Metrics storage
Certification engine
Pilot engine

Determine:

Can the system currently produce:

- Win Rate
- Profit Factor
- Sharpe
- Sortino
- Expectancy
- Drawdown
- Regime statistics

from real recorded trades?

If not:

Show exactly what is missing.

Build the highest ROI roadmap possible.

Rank every remaining improvement by:

Expected Profit Impact
Expected Win Rate Impact
Expected PF Impact
Expected Sharpe Impact
Implementation Effort

Format:

Priority | Task | Effort | Expected Impact | ROI Score

Determine whether the current strategy inventory is sufficient.

Answer:

1. Does the application already contain enough alpha to become profitable?
2. If all dormant alpha modules are activated, is additional strategy development required?
3. What percentage of expected profitability can be achieved using existing strategies only?
4. What alpha sources are completely missing?
5. What new institutional-grade strategies should be added after existing alpha is activated?

Rank the Top 20 future strategy additions by expected value.

Provide a hedge-fund style executive conclusion.

Answer explicitly:

- A. Is there evidence that profitable alpha exists in the codebase?
- B. Is the current system trading its strongest strategies?
- C. What percentage of the strongest alpha is currently active?
- D. What is the estimated ceiling after activating dormant alpha?
- E. What is the single highest ROI action remaining?
- F. Should new strategies be built now, or should existing alpha be activated first?
- G. What would you do if you were deploying institutional capital into this system?

Use only code evidence.

Do not guess.

Do not estimate profitability without supporting implementation evidence.

Provide exact file names, functions, execution paths, and technical proof for every conclusion.

This prompt will force Claude to perform a full hedge-fund-level audit and tell you whether the existing dormant alpha is enough, or whether you truly need new strategies.



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Phase 22B HealthScore Audit

There are two separate HealthScore concepts — one in the risk/strategy path, one in SOR/venue path. The audit covers both.

1. Strategy HealthScore (0–100)

Generation

File Function Line What it does

strategy_tracker.go trackerHealthScore() 534–568

Computes 0–100 from WinRate, ProfitFactor,
MaxDrawdown, ConsecutiveLosses, TotalTrades



PHASE 22E — INSTITUTIONAL STATISTICAL CERTIFICATION, PROFITABILITY VALIDATION & CAPITAL DEPLOYMENT FRAMEWORK

You are the Lead Quant Researcher, Head of Systematic Trading, Institutional Portfolio Manager, Statistical Validation Specialist, Quant Risk Officer, Backtesting Architect, and Independent Hedge Fund Auditor.

Your mission is to execute a COMPLETE Phase 22E implementation across the entire trading platform.

IMPORTANT:

Do not estimate.

Do not fabricate.

Do not assume profitability.

Do not infer results from architecture quality.

Do not use hypothetical metrics.

Every conclusion must be derived from actual trade records, actual backtests, actual paper trades, actual fills, or actual statistical calculations.

A strategy is NOT profitable because it looks good.

A strategy is profitable only if the data proves it.

Determine with statistical evidence:

1. Which strategies are profitable.
2. Which alpha engines are profitable.
3. Which strategies should receive capital.
4. Which strategies should be disabled.
5. Which strategies should be retired permanently.
6. Which strategies deserve live deployment.
7. Whether the system has genuine edge.

Generate a complete institutional certification framework.

First verify:

Do sufficient trade records exist?

Audit all locations:

Paper trades

Backtest trades

Walk-forward trades

Live fills

MongoDB

PostgreSQL

Event store

Audit logs

Trade projections

Strategy tracker records

Generate:

DATA_CERTIFICATION_AUDIT.md

For every source report:

Trade count

Date range

Strategies covered

Missing fields

Data quality score

If data is insufficient:

Document exact gaps.

Do not continue certification using fabricated data.

Inventory every strategy.

Generate:

STRATEGY_REGISTRY_AUDIT.md

For each strategy:

Name

Category

Family

Alpha Source

Version

Status

Active/Disabled

Trade Count

Classify:

Institutional Alpha

Order Flow

Smart Money

Funding

Liquidation

Liquidity

Mean Reversion

Momentum

Trend

Indicator-Based

Expansion Pack

Experimental

Produce complete rankings.

Select:

Top 20 strategies

Based on:

Execution volume

Signal quality

Alpha ranking

Recent activity

Run:

500 completed trades minimum per strategy

For each strategy compute:

Total Trades

Win Rate

Loss Rate

Profit Factor

Gross Profit

Gross Loss

Net PnL

Average Win

Average Loss

Expectancy

Sharpe

Sortino

Calmar

Recovery Factor

Max Drawdown

Average Holding Time

Longest Losing Streak

Longest Winning Streak

Generate:

TOP20_VALIDATION_REPORT.md

Select:

Top 10 performers from Phase 3

Target:

1000+ completed trades per strategy

Compute:

All metrics above plus:

Confidence Intervals

Variance

Standard Error

Skewness

Kurtosis

Trade Distribution

Return Distribution

Generate:

TOP10_CERTIFICATION_REPORT.md

For every certified strategy:

Run:

1000 Monte Carlo simulations

Measure:

Expected Return

Worst Case Return

Best Case Return

Expected Drawdown

Worst Drawdown

Probability Of Ruin

Probability Of Capital Growth

Generate:

MONTE_CARLO_REPORT.md

Classify:

Stable

Marginal

Unstable

Untradable

Evaluate performance separately in:

Bull Markets

Bear Markets

Range Markets

High Volatility

Low Volatility

Funding Positive

Funding Negative

High Liquidation Events

Low Liquidation Events

Strong Trend

Weak Trend

Measure:

PF

Sharpe

Expectancy

Drawdown

Generate:

REGIME_ANALYSIS_REPORT.md

Identify:

Strategies dependent on specific regimes.

Evaluate separately:

Funding Mean Reversion

Liquidation Cascade

Liquidity Sweep

Fair Value Gap

Order Block

Market Structure Shift

Session Expansion

Order Flow

Market Profile

Statistical Mean Reversion

For each alpha source:

Trades

Win Rate

Profit Factor

Sharpe

Expectancy

Drawdown

Generate:

ALPHA_CERTIFICATION_REPORT.md

Rank all alpha engines.

Measure:

Strategy-to-strategy correlation

Family correlation

Alpha correlation

Regime correlation

Create:

Correlation matrix

Cluster map

Diversification score

Generate:

PORTFOLIO_CORRELATION_REPORT.md

Identify:

Redundant strategies

Duplicate strategies

Overlapping alpha

Automatically identify strategies that should be retired.

Retirement conditions:

Profit Factor < 1.00

Negative Expectancy

Sharpe < 0.50

Persistent drawdown

Unstable Monte Carlo

No statistical edge

Generate:

RETIREMENT_CANDIDATES_REPORT.md

Rank by severity.

Create institutional ranking.

Score using:

Profit Factor 30%

Sharpe 25%

Expectancy 20%

Drawdown 15%

Stability 10%

Generate:

CAPITAL_DEPLOYMENT_REPORT.md

For every strategy recommend:

0%

5%

10%

15%

20%

25%

capital allocation.

Classify every strategy:

NOT CERTIFIED

PAPER ONLY

PILOT CAPITAL

LIMITED CAPITAL

FULL DEPLOYMENT

INSTITUTIONAL GRADE

Certification requirements:

NOT CERTIFIED

$PF < 1.00$

PAPER ONLY

$PF \geq 1.10$

PILOT CAPITAL

$PF \geq 1.20$

$\text{Sharpe} \geq 1.00$

LIMITED CAPITAL

$PF \geq 1.30$

Sharpe ≥ 1.25

FULL DEPLOYMENT

PF ≥ 1.40

Sharpe ≥ 1.50

INSTITUTIONAL GRADE

PF ≥ 1.50

Sharpe ≥ 2.00

1000+ trades

Max DD < 10%

Generate:

LIVE_DEPLOYMENT_CERTIFICATION.md

Answer explicitly:

Does the system have edge?

Which strategies have edge?

Which alpha engines have edge?

What is the expected annual return?

What is the expected drawdown?

What is the expected portfolio Sharpe?

What percentage of strategies are profitable?

What percentage should be retired?

What percentage deserve capital?

Support every answer with actual statistical evidence.

Build:

Certification Dashboard

Expose:

Strategy Rankings

Alpha Rankings

Profit Factor Rankings

Sharpe Rankings

Drawdown Rankings

Capital Recommendations

Certification Status

Prometheus Metrics

JSON Endpoints

Historical Trends

Create:

Unit Tests

Integration Tests

Statistical Validation Tests

Monte Carlo Tests

Certification Tests

Portfolio Tests

Regression Tests

Verify:

All calculations

All rankings

All classifications

All reports

All certification decisions

Generate:

PHASE_22E_IMPLEMENTATION_REPORT.md

Include:

1. Data availability audit
2. Strategy inventory
3. Top 20 validation results
4. Top 10 certification results
5. Monte Carlo findings
6. Regime analysis
7. Alpha rankings
8. Correlation analysis
9. Retirement candidates
10. Capital allocation recommendations
11. Live deployment recommendations
12. Proven profitability assessment
13. Remaining risks
14. Production readiness score

No strategy receives capital without evidence.

No strategy is certified without statistics.

No alpha engine is approved without validation.

All rankings must be reproducible.

All calculations must be traceable.

All conclusions must be evidence-backed.

No estimates.

No assumptions.

No fabricated results.

Produce a complete hedge-fund-grade profitability certification framework that determines, with statistical evidence, whether the application possesses genuine trading edge and exactly which strategies deserve capital deployment.

The final report must allow an independent quantitative researcher to reproduce every conclusion and verify every profitability claim

from raw trade data through final capital allocation decisions.

22F → 1000 Trade Validation Generate a highly detailed, expert-level, single comprehensive prompt that can be directly pasted into Claude Agent to perform this task with maximum accuracy, depth, completeness, and production-grade quality

:::writing{variant="document" id="67184"} PHASE 22F — INSTITUTIONAL 1000-TRADE VALIDATION, EDGE VERIFICATION & PROFITABILITY PROOF ENGINE You are the Chief Quantitative Research Officer, Head of Systematic Trading, Institutional Portfolio Manager, Quant Statistician, Monte Carlo Specialist, Risk Committee Chair, and Independent Hedge Fund Auditor. Your mission is to perform a COMPLETE Phase 22F implementation whose sole objective is to determine whether the trading system possesses statistically valid, repeatable, deployable trading edge. IMPORTANT: Do not assume profitability. Do not infer profitability from architecture quality. Do not infer profitability from code quality. Do not infer profitability from backtest screenshots. Do not estimate metrics. Do not fabricate statistics. Only use actual trade records, actual fills, actual backtests, actual paper trading results, actual walk-forward outputs, and actual event data. No strategy is profitable until the data proves it.

=====

===== PRIMARY OBJECTIVE

=====

===== Prove or disprove: 1. Does the system have edge? 2. Which strategies have edge? 3. Which alpha engines have edge? 4. Which strategies survive 1000+ trade validation? 5. Which strategies deserve capital? 6. Which strategies must be retired? 7. Can the portfolio be deployed safely? Generate a complete institutional-grade profitability certification system.

=====

===== PHASE 1 — DATA INTEGRITY CERTIFICATION

=====

===== Audit every data source. Inventory: Paper Trades Backtest V2 Backtest V3 Walk Forward Monte Carlo MongoDB PostgreSQL Event Store OMS Execution Intelligence Trade Conversion Certification Engine Verify: Trade counts Duplicate records Missing records Corrupted records Survivorship bias Look-ahead bias Data leakage Generate: DATA_INTEGRITY_CERTIFICATION.md Produce: Certified Trade Count Certified Fill Count Certified Strategy Count Only certified records may proceed.

```
=====
===== PHASE 2 — TOP STRATEGY SELECTION
=====
===== Using actual evidence select: Top 20 strategies. Ranking inputs: Profit
Factor Sharpe Expectancy Execution Quality Signal Quality Alpha Quality Drawdown Trade
Count Stability Create: TOP20_SELECTION_REPORT.md Justify every selection.
=====
===== PHASE 3 — 1000 TRADE VALIDATION CAMPAIGN
=====
===== For each Top 20 strategy: Target: 1000 completed trades minimum Do
not stop at: 100 200 500 750 Continue until: 1000 completed trades or strategy invalidated
Track: Trade Number PnL Equity Curve Drawdown Regime Execution Quality Signal Quality
Store: Full trade history Generate: STRATEGY_VALIDATION_DATASET
=====
===== PHASE 4 — COMPLETE STATISTICAL ANALYSIS
=====
===== For every strategy compute: Total Trades Win Rate Loss Rate Breakeven
Rate Profit Factor Gross Profit Gross Loss Net Profit Expectancy Average Win Average Loss
Reward Risk Ratio Sharpe Ratio Sortino Ratio Calmar Ratio Recovery Factor Ulcer Index
Maximum Drawdown Average Drawdown Trade Duration Consecutive Wins Consecutive
Losses Risk Of Ruin Probability Of Profitability Generate:
STATISTICAL_VALIDATION_REPORT.md
=====
===== PHASE 5 — CONFIDENCE INTERVALS
=====
===== For every metric calculate: 90% 95% 99% confidence intervals
Specifically: Profit Factor Sharpe Expectancy Win Rate Determine: Statistical significance
Margin of error Reliability score Generate: CONFIDENCE_ANALYSIS_REPORT.md
=====
===== PHASE 6 — MONTE CARLO STRESS TESTING
=====
===== Run: 1000 simulations per strategy Measure: Expected Return Worst
Return Best Return Expected Drawdown Worst Drawdown Risk Of Ruin Capital Survival
Rate Tail Risk Generate: MONTE_CARLO_CERTIFICATION.md Classify: Robust Stable
Marginal Unstable Failed
=====
===== PHASE 7 — REGIME PERFORMANCE CERTIFICATION
=====
===== Separate trades by: Bull Bear Range Volatile Low Volatility High Funding
Negative Funding Liquidation Events Trend Mean Reversion For every regime calculate: PF
Sharpe Expectancy Drawdown Generate: REGIME_CERTIFICATION.md Identify: Regime-
dependent strategies.
```

```
=====
===== PHASE 8 — ALPHA ENGINE VALIDATION
=====
===== Validate individually: Funding Mean Reversion Liquidation Cascade
Liquidity Sweep Fair Value Gap Order Block Market Structure Shift Session Expansion Order
Flow Market Profile Statistical Mean Reversion For each alpha: Trades Win Rate Profit
Factor Sharpe Expectancy Max Drawdown Execution Quality Generate:
ALPHA_EDGE_REPORT.md Rank strongest alpha sources.
=====
===== PHASE 9 — EXECUTION QUALITY CORRELATION
=====
===== Using Phase 22D data: Correlate: Latency Slippage Missed Entries Fill
Quality TP Overrides Signal Age Against: Profit Factor Sharpe Expectancy Determine: How
much execution quality affects profitability. Generate:
EXECUTION_CORRELATION_REPORT.md
=====
===== PHASE 10 — PORTFOLIO CONSTRUCTION
=====
===== Build: Optimal Portfolio Use: Correlation Drawdown Sharpe Profit
Factor Expectancy Monte Carlo Stability Construct: Top 5 Top 10 Top 20 Portfolio Variants
Generate: PORTFOLIO_OPTIMIZATION_REPORT.md
=====
===== PHASE 11 — CAPITAL ALLOCATION CERTIFICATION
=====
===== Allocate capital only from validated evidence. Weighting: Profit Factor
30% Sharpe 25% Expectancy 20% Drawdown 10% Stability 10% Execution Quality 5%
Generate: CAPITAL_DEPLOYMENT_CERTIFICATION.md Recommend: 0% 5% 10% 15% 20%
25% allocation bands.
=====
===== PHASE 12 — STRATEGY ELIMINATION ENGINE
=====
===== Automatically identify: Strategies to disable. Conditions: PF < 1.00
Negative Expectancy Sharpe < 0.50 Monte Carlo Failure Risk Of Ruin Excessive Persistent
Drawdown Generate: STRATEGY_RETIREMENT_REPORT.md Provide exact reasoning.
=====
===== PHASE 13 — INSTITUTIONAL CERTIFICATION TIERS
=====
===== Classify: FAILED WATCHLIST PAPER ONLY PILOT LIMITED CAPITAL FULL
DEPLOYMENT INSTITUTIONAL GRADE Requirements: FAILED PF < 1.00 WATCHLIST PF ≥
1.00 PAPER ONLY PF ≥ 1.10 PILOT PF ≥ 1.20 Sharpe ≥ 1.00 LIMITED PF ≥ 1.30 Sharpe ≥
1.25 FULL DEPLOYMENT PF ≥ 1.40 Sharpe ≥ 1.50 INSTITUTIONAL PF ≥ 1.50 Sharpe ≥ 2.00
1000+ trades Max DD < 10% Generate: INSTITUTIONAL_CERTIFICATION_REPORT.md
```

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===== PHASE 14 — EDGE VERDICT

=====

===== Answer explicitly: Does the system possess edge? What is the strongest strategy? What is the strongest alpha engine? What is the expected portfolio PF? What is the expected portfolio Sharpe? What is the expected drawdown? How many strategies passed? How many failed? What percentage deserve capital? What percentage should be retired? Support every answer with actual statistics.

=====

===== PHASE 15 — AUTOMATED VALIDATION PIPELINE

=====

===== Build automation: Every new strategy must automatically: Generate trades Reach minimum sample size Run certification Run Monte Carlo Run regime analysis Receive deployment grade No strategy may reach capital deployment without passing validation.

=====

===== PHASE 16 — DASHBOARD & OBSERVABILITY

=====

===== Expose: Validation Status Trade Counts PF Rankings Sharpe Rankings Expectancy Rankings Alpha Rankings Certification Tiers Capital Recommendations Prometheus Metrics REST Endpoints Historical Trends

=====

===== PHASE 17 — TESTING

=====

===== Create: Unit Tests Integration Tests Validation Tests Monte Carlo Tests Portfolio Tests Certification Tests Regression Tests Verify: All calculations All rankings All classifications All reports All certification decisions

=====

===== FINAL DELIVERABLE

=====

===== Generate: PHASE_22F_IMPLEMENTATION_REPORT.md Containing: 1. Data Integrity Certification 2. Top 20 Strategy Selection 3. 1000 Trade Validation Results 4. Statistical Analysis 5. Confidence Interval Analysis 6. Monte Carlo Results 7. Regime Analysis 8. Alpha Rankings 9. Execution Correlation Findings 10. Portfolio Construction Results 11. Capital Deployment Recommendations 12. Retirement Candidates 13. Certification Tiers 14. Edge Verdict 15. Production Readiness Assessment

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===== MANDATORY RULES

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===== No assumptions. No fabricated metrics. No estimated profitability. No deployment recommendations without evidence. Every profitability claim must be traceable to actual trade records. Every certification decision must be reproducible. Every

ranking must be statistically justified. The final output must be sufficient for an institutional investment committee to decide whether the system deserves live capital deployment. :::

Phase 22F is complete. Build: PASS. Tests: 17/17 PASS.

What was built

Package: engine/internal/validation/phase22f/ (19 files)

Phase File Delivers

- 1 data_integrity.go Audits all data sources — dedup, look-ahead bias, survivorship bias, certified trade count
- 2 top20.go Evidence-based top-20 selection via 7-criteria composite score
- 3 campaign.go 1000-trade validation campaign with milestones at 100/200/500/750/1000; auto-invalidates if

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